

```
{  
  token: 112,  
  userRole: "admin"  
}
```

Options :

6406534219691. ✖ You are not logged in.
6406534219692. ✔ Welcome to profile
6406534219693. ✖ You are not authorized.
6406534219694. ✖ None of these.

MLT

Section Id :	64065387652
Section Number :	7
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	17
Number of Questions to be attempted :	17
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653194652
Question Shuffling Allowed :	No

Question Number : 153 Question Id : 6406531252597 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING TECHNIQUES (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE [TOP](#) FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534219695. ✓ YES

6406534219696. ✗ NO

Sub-Section Number :

2

Sub-Section Id :

640653194653

Question Shuffling Allowed :

Yes

Question Number : 154 Question Id : 6406531252598 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

What is the role of the regularization term in ridge regression?

Options :

6406534219697. ✗ It ensures that the dataset is normally distributed.

6406534219698. ✓ It penalizes large coefficients to reduce overfitting.

6406534219699. ✗ It forces more coefficients to be exactly zero, performing feature selection.

6406534219700. ✗ It increases the variance of the model.

Question Number : 155 Question Id : 6406531252605 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following conditions must be satisfied for a decision tree to be pure at a given node?

Options :

6406534219723. ✗ The entropy of the node is maximum.

6406534219724. ✗ The information gain at the node is negative.

6406534219725. ✓ All samples at a node belong to the same class.

6406534219726. ✓ The entropy of the node is 0.

Sub-Section Number :

3

Sub-Section Id :

640653194654

Question Shuffling Allowed :

Yes

Question Number : 156 Question Id : 6406531252599 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider that the three weight vectors w_1, w_2 , and w_3 are learned for a four-dimensional dataset using a linear regression model or regularized linear regression model (Not in any particular order).

$$w_1 = [-0.41 \ 0.26 \ 0.54 \ 0.17]^T$$

$$w_2 = [-0.81 \ 0.89 \ 0.93 \ 0.52]^T$$

$$w_3 = [0 \ 0 \ 0.08 \ 0.2]^T$$

Select the most appropriate match for these weight vectors.

Options :

6406534219701. ✖ $w_1 \rightarrow$ Linear regression, $w_2 \rightarrow$ Ridge regression, $w_3 \rightarrow$ Lasso regression

6406534219702. ✔ $w_1 \rightarrow$ Ridge regression, $w_2 \rightarrow$ Linear regression, $w_3 \rightarrow$ Lasso regression

6406534219703. ✖ $w_1 \rightarrow$ Lasso regression, $w_2 \rightarrow$ Ridge regression, $w_3 \rightarrow$ Linear regression

6406534219704. ✖ $w_1 \rightarrow$ Lasso regression, $w_2 \rightarrow$ Linear regression, $w_3 \rightarrow$ Ridge regression

Question Number : 157 Question Id : 6406531252602 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following dataset in $\mathbb{R}^2$:

$$\{(1, 2), (3, 3), (3, 4), (5, 4), (7, 8), (9, 10)\}$$

Suppose we want to form 3 clusters from these points using Lloyd's algorithm. After the first iteration, clusters C_1, C_2 , and C_3 have the following data points:

$$C_1 : \{(1, 2), (3, 3)\}$$

$$C_2 : \{(3, 4), (5, 4)\}$$

$$C_3 : \{(7, 8), (9, 10)\}$$

After second iteration, which of the clusters is the data point $(3, 3)$ expected to move to?

Options :

6406534219711. ✔ C_1

6406534219712. ✖ C_2

6406534219713. ✖ C_3

Sub-Section Id :

640653194655

Question Shuffling Allowed :

Yes

Question Number : 158 Question Id : 6406531252600 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

A dataset of 1000 second-hand cars has four features: kilometers driven (x_1), mileage (x_2), the present price of the car (x_3), and the selling price (x_4). The selling price seems to have the following relationship (approximate) with the other three features.

$$x_4^2 = x_1x_2 + x_1x_2x_3.$$

If we want to project the dataset into a lower dimensional space, which of the following task would be most appropriate?

Options :

6406534219705. ✖ Kernel PCA with a polynomial kernel of degree 2.

6406534219706. ✔ Kernel PCA with a polynomial kernel of degree 3.

6406534219707. ✖ Kernel PCA with a polynomial kernel of degree 4.

6406534219708. ✖ Standard PCA

Sub-Section Number :

5

Sub-Section Id :

640653194656

Question Shuffling Allowed :

Yes

Question Number : 159 Question Id : 6406531252601 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Weak learners used in the ensemble method typically perform slightly better than random guessing.

Options :

6406534219709. ✔ TRUE

6406534219710. ✖ FALSE

Sub-Section Number :

6

Sub-Section Id :

640653194657

Question Shuffling Allowed :

Yes

Question Number : 160 Question Id : 6406531252603 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

A set of data points is generated by the following process:

$$y_i = w_0 + w_1x_i + w_2x_i^2 + w_3x_i^3 + \epsilon, \text{ where } \epsilon \text{ is a Gaussian noise.}$$

We use two models to fit the data:

Model 1: $\hat{y} = a_0 + a_1x$

Model 2: $\hat{y} = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$.

For a fixed number of training examples, select the correct options from the following:

Options :

6406534219714. ✓ Model 1 will have a higher bias than Model 2.

6406534219715. ✗ Model 2 will have a lower variance than Model 1.

6406534219716. ✓ Model 1 is more likely to underfit.

6406534219717. ✗ Model 1 is more likely to overfit.

Question Number : 161 Question Id : 6406531252604 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let w^*, ξ^* be the optimal primal solutions, and α^*, β^* be the optimal dual solutions of the soft-margin SVM problem. Select the options which are always true.

Options :

6406534219718. ✗ If $\alpha_i^* = C$, the i^{th} point is incorrectly classified by w^* .

6406534219719. ✓ If $\alpha_i^* = 0$, the i^{th} point is correctly classified by w^* .

6406534219720. ✗ If the i^{th} data point lies on one of the supporting hyperplanes, then $\alpha_i^* = 0$.

6406534219721. ✓ If the i^{th} data point lies on one of the supporting hyperplanes, it does not pay any bribes.

6406534219722. ✓ If the i^{th} data point lies between the supporting hyperplanes, it always pay a positive bribe.

Sub-Section Number :

7

Sub-Section Id :

640653194658

Question Shuffling Allowed :

Yes

Question Number : 162 Question Id : 6406531252612 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the following data set with three data points:

$$D = \left\{ \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, +1 \right), \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}, -1 \right), \left(\begin{bmatrix} -1 \\ 0 \end{bmatrix}, -1 \right) \right\}.$$

The perceptron algorithm is applied to the dataset with an initial weight vector $w^0 = [0 \ 0]^T$. While looking for mistakes, cycle through the data points from left to right. Which of the following statements are true?

Options :

6406534219733. ✖ The algorithm will converge with $w = [1 \ 1]^T$.

6406534219734. ✔ The algorithm will converge with $w = [1 \ 2]^T$.

6406534219735. ✖ The algorithm will converge with $w = [0 \ 2]^T$.

6406534219736. ✖ The algorithm will converge with $w = [0 \ 1]^T$.

6406534219737. ✔ The model will predict the label as +1, for the new data point $[2 \ -1]^T$.

6406534219738. ✖ The model will predict the label as -1, for the new data point $[2 \ -1]^T$.

Sub-Section Number :

8

Sub-Section Id :

640653194659

Question Shuffling Allowed :

Yes

Question Number : 163 Question Id : 6406531252606 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

A kernel regression model with a polynomial kernel of degree 2 is trained on the following dataset with three features:

$$X = \begin{bmatrix} 1 & 4 & 7 & 2 & 6 \\ 2 & 5 & 8 & 1 & 5 \\ 3 & 6 & 9 & 3 & 4 \end{bmatrix}.$$

Let the optimal weight vector w^* be given by:

$$w^* = \phi(X) [1, 1.2, 0, 0.7, -0.5]^T,$$

where ϕ is the transformation mapping corresponding to the kernel $K(x, x') = (x^T x' + 1)^2$. Compute the predicted value y_{pred} for a new data point $x' = [1 \ 1 \ 1]^T$. Enter the answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

262.2 to 262.8

Question Number : 164 **Question Id :** 6406531252607 **Question Type :** SA

Correct Marks : 4

Question Label : Short Answer Question

A coin is flipped 1000 times, and it lands on heads 150 times. You use a Beta(10, 40) prior to estimate the true probability of heads. Given these observations, calculate the posterior mean of the probability of heads. Enter the answer correct to three decimal places.

Note:

- Pdf of Beta distribution is $f(\theta; \alpha, \beta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)}$, $0 \leq \theta \leq 1, \alpha, \beta \geq 0$, where $B(\alpha, \beta)$ is the Beta function.
- The mean of the Beta distribution is: $E[\theta] = \frac{\alpha}{\alpha+\beta}$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.148 to 0.154

Question Number : 165 Question Id : 6406531252609 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

Consider a single-layer neural network comprising two neurons in the hidden layer. The parameters are given as below:

$$w_1^{(1)} = \frac{1}{2}, \quad w_2^{(1)} = \frac{1}{2}$$

$$w_1^{(2)} = 1, \quad w_2^{(2)} = -1$$

$$w^{out} = [2 \quad -2]^T,$$

where $w_i^{(j)}$ represents the weight associated with the j -th neuron for the i -th data point. Suppose we are solving the regression problem, then determine the output of the neural network using RELU as the activation function for the input $x = [1 \quad 3]^T$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4

Question Number : 166 Question Id : 6406531252611 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

The following dataset represents different weather conditions and the corresponding decision on whether a person chooses to play tennis:

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Overcast	Hot	High	Weak	No
2	Sunny	Hot	High	Weak	No
3	Sunny	Hot	Low	Weak	Yes
4	Overcast	Hot	High	Weak	Yes
5	Overcast	Cool	Low	Strong	Yes
6	Sunny	Cool	High	Weak	Yes
7	Sunny	Cool	Low	Weak	No
8	Sunny	Cool	Low	Strong	No
9	Sunny	Hot	Low	Strong	Yes
10	Overcast	Hot	High	Weak	Yes

Using the Naïve Bayes algorithm, compute the probability that a person will play tennis (Yes) given the conditions:

- Outlook = Overcast
- Temperature = Cool
- Humidity = High
- Wind = Weak

Enter the answer up to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.61 to 0.67

Sub-Section Number : 9

Sub-Section Id : 640653194660

Question Shuffling Allowed : Yes

Question Number : 167 **Question Id :** 6406531252608 **Question Type :** SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a logistic regression model that has been trained for a binary classification problem on a dataset in $\mathbb{R}^2$. Given a test data-point as input to the model, it returns 1 as the predicted label if the probability output by the model is greater than 0.55 and 0 otherwise. What will be the predicted label for the test data-point $\begin{bmatrix} 1 & 2 \end{bmatrix}^T$? Assume the weight vector is $\begin{bmatrix} 7 & 4 \end{bmatrix}^T$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

Question Number : 168 Question Id : 6406531252610 Question Type : SA
Correct Marks : 3

Question Label : Short Answer Question

Consider the following training dataset for a binary classification problem in $\mathbb{R}^2$. Each data-point is represented by $X = [x_1 \ x_2]^T$ with label y .

x_1	x_2	y
1	0	1
-1	1	-1
5	4	1
-5	-4	-1

A hard-margin linear-SVM is trained on this dataset. If the solution to the dual problem is

$$\alpha^* = \begin{bmatrix} 0.1 \\ 0.4 \\ 0 \\ 0 \end{bmatrix},$$

then find the prediction for the data point $x_{test} = [1 \ -2]^T$?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

Sub-Section Number : 10
Sub-Section Id : 640653194661
Question Shuffling Allowed : No

Question Id : 6406531252613 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (169 to 170)
Question Label : Comprehension

Consider a single iteration of the AdaBoost algorithm that is run on five sample points $\{x_1, x_2, x_3, x_4, x_5\}$ with labels either 1 or -1 , starting with uniform weights. Two classifiers h_1 and h_2 predict the labels as shown in the below table:

Data point	True label	Predicted label(h_1)	Predicted label(h_2)
x_1	1	-1	1
x_2	-1	1	1
x_3	-1	-1	1
x_4	1	1	-1
x_5	-1	-1	1

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 169 Question Id : 6406531252614 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which classifier has the lower training error?

Options :

6406534219739. ✓ h_1

6406534219740. ✗ h_2

Question Number : 170 Question Id : 6406531252615 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The classifier with the lower training error rate is used in the next round of the AdaBoost algorithm. Determine the classifier h^* obtained at the end of this round.

Options :

6406534219741. ✗ $h^*(x) = \text{sign}(0.202 * h_2)$

6406534219742. ✓ $h^*(x) = \text{sign}(0.202 * h_1)$

6406534219743. ✗ $h^*(x) = \text{sign}(-0.69 * h_1)$

6406534219744. ✖ $h^*(x) = \text{sign}(-0.69 * h_2)$

MLP

Section Id :	64065387653
Section Number :	8
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	27
Number of Questions to be attempted :	27
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653194662
Question Shuffling Allowed :	No

Question Number : 171 Question Id : 6406531252616 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

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Options :

6406534219745. ✔ YES

6406534219746. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653194663
Question Shuffling Allowed :	Yes